MOPA Fiber Metal Relief Engraving

ZoeDepth to LightBurn 3D Sliced Workflow for an OMTech 60W MOPA

Optimized for metal relief, coins, medallions, and bas-relief style engraving

Core takeaway: ZoeDepth is a useful first-pass depth estimator, but the final LightBurn map must be edited into a bas-relief heightmap. For 60W MOPA metal work, use LightBurn 3D Sliced, calibrate black-to-depth with test strips, and rely on cleanup passes for finish quality.

1. Executive Summary

- **Correct machine category:** A 60W MOPA fiber/galvo is the correct type of laser for real metal relief engraving. This is different from CO2 metal marking, which usually requires coatings or marking spray.
- **Correct LightBurn mode:** Use Image mode with 3D Sliced for depth maps. In 3D Sliced, darker pixels receive more passes and engrave deeper; enabling Negative Image reverses that behavior.
- **Correct depth-map mindset:** ZoeDepth gives camera-estimated depth, not a clean bas-relief. Use it as the starting map, then manually flatten, exaggerate, smooth, and repaint depth planes.
- **Correct testing mindset:** There is no universal value for "black equals X mm deep." Actual depth depends on metal, lens, focus, speed, power, frequency, pulse width, interval, pass count, and cleanup strategy.

2. Machine and Project Assumptions

- Laser: OMTech 60W MOPA fiber/galvo, preferably JPT MOPA or equivalent pulse-width-adjustable source.
- Software: LightBurn with galvo support and access to 3D Sliced Image Mode.
- Primary materials: brass, aluminum, stainless steel, and possibly copper.
- Desired output: actual physical relief/depth in metal, not merely dark photo marking.
- Map input: grayscale heightmap/depth map generated from ZoeDepth or another AI depth model, then manually edited.

3. What Other People Are Doing

- Using AI depth generators such as DepthR, marcam.uk, SculptOK, Tiled ZoeDepth, and ZoeDepth/Depth Anything based tools to make first-pass depth maps.
- Engraving brass coins and medallions with 60W MOPA fiber lasers in LightBurn 3D Sliced mode.
- Starting around 256 slices/passes for practical tests, then moving to 512 for higher-detail final pieces when runtime is acceptable.
- Editing the AI depth map heavily in Photoshop, GIMP, Krita, Affinity, or similar tools before engraving.
- Using cleanup passes to remove debris and improve surface finish, especially on stainless and deeper brass work.

Important community lesson: AI depth estimation is not the final art. Successful users commonly generate a basic depth map, then blend overlays, repaint bad zones, simplify backgrounds, and manually adjust tonal depth before using 3D Slice.

4. Recommended End-to-End Workflow

```
Source photo or AI art
-> ZoeDepth / Tiled ZoeDepth depth estimation
-> Normalize and invert depth
-> Flatten or remove background
-> Manually edit into bas-relief planes
-> Smooth noisy texture but preserve major edges
-> Export grayscale PNG
-> LightBurn Image Mode: 3D Sliced
-> Test strip / tone calibration
-> Full engrave with cleanup passes
-> Polish / brush / clean final piece
```

5. Preparing the ZoeDepth Map for Metal Relief

For LightBurn 3D Sliced, the heightmap should describe intended engraving depth, not photographic shading. A clean relief map usually has fewer, stronger tonal relationships than a raw depth estimate.

5.1 Target grayscale meaning

Tone	Meaning in 3D Sliced
White / near-white	Top surface / shallowest area / fewest passes
Mid-gray	Intermediate relief planes
Dark gray	Deeper forms and recesses
Black / near-black	Deepest cuts; use sparingly

5.2 Suggested ZoeDepth processing values

Processing step	Starting value / rule
Clip depth range	5th to 95th percentile, then adjust by subject
Invert	Yes, so closer subject becomes darker/deeper if using default 3D Sliced behavior
Gamma	0.65 to 0.85; lower gamma exaggerates midtone relief
Background	Push toward white / shallow; avoid wasting passes on noisy background
Smoothing	Light bilateral or Gaussian smoothing; avoid destroying major edges
Sharpening	Use very lightly and late; edge halos can engrave as unwanted ridges
Export	Clean grayscale PNG; keep a high-bit master and an 8-bit LightBurn-ready copy

5.3 Manual edits that usually matter most

- **Flatten the background:** Make the subject the only area with meaningful depth. Background depth makes metal relief muddy and slow.

- **Repaint thin details:** Hair, chains, small text, outlines, eyes, ears, and reflective areas often need hand correction.
- **Create intentional planes:** Good bas-relief reads as clear layers. Raw depth maps often look soft and uncertain.
- **Reserve black:** Do not make half the image black. Black means maximum number of passes and runtime.
- **Check at engraving size:** Tiny gradients that look good on screen may disappear or become rough at coin scale.

6. LightBurn 3D Sliced Setup

1. Import the processed grayscale depth map into LightBurn.
2. Set the layer to Image mode.
3. Choose 3D Sliced as the Image Mode.
4. Confirm black/deep and white/shallow behavior. If the result is reversed, use Negative Image or invert the source map.
5. Set your passes/slices. Start with 128 for short tests, 256 for first real attempts, and 512 only after the map is proven.
6. Use Angle Increment so repeated passes do not all follow the same scan direction.
7. Enable cleanup passes if debris, roughness, or darkness builds up.

7. Starting Settings for 60W MOPA on Brass

These are starting points only. They are based on common 60W MOPA brass/coin workflows and must be tested on your exact metal, lens, focus, and desired depth.

Parameter	Conservative start	More aggressive / final
Lens	150mm	150mm or 110mm for smaller/detail work
Speed	1800-2200 mm/s	2000-3000 mm/s depending on depth strategy
Power	85-92%	90-95%
Frequency	75-105 kHz	100-120 kHz if finish improves
Pulse width / Q-pulse	100-150 ns	150-200 ns
Line interval	0.025 mm	0.020-0.025 mm
Passes / slices	128 test, 256 first real attempt	512 final if worthwhile
Scan angle	0 or 45 degrees	Use Angle Increment 45 or 60 degrees
Focus	Exact surface focus first	Test tiny defocus only after baseline

8. Cleanup Pass Strategy

Cleanup passes are especially useful on metal relief because re-deposited material, oxidation, and rough edges can obscure depth. LightBurn describes cleanup passes as generally using a tighter interval, higher frequency, and lower power than the main engraving passes.

Cleanup setting	Starting value
Clean every	10-25 layers for deeper brass/stainless tests
Cleanup speed	2500-3500 mm/s

Cleanup power	25-40%
Cleanup frequency	80-120 kHz
Cleanup pulse width	100-200 ns
Cleanup interval	0.015-0.025 mm

9. Material-Specific Notes

Material	Behavior	Practical advice
Brass	Best starting metal for coin-style relief. Good contrast and depth potential.	Start with 256 slices, 0.025 mm interval, 90-95% power, 75-105 kHz, 100-200 ns.
Aluminum	Often cuts/marks cleaner than stainless; may lack the rich look of brass.	Use cleanup passes, test higher frequency, and avoid excessive heat buildup.
Stainless steel	Debris and heat effects can build up; surface can get rough.	Use lower power per pass, more passes, frequent cleanup, and do not brute-force depth too quickly.
Copper	Reflective and slower; can blacken and may require long runtimes.	Expect more testing. Reduce ambitions for first tests and use careful cleanup/polishing.

10. Calibration Test Procedure

Before engraving a full piece, create a small grayscale ramp that maps tone to real engraved depth. This prevents wasting hours on a full-size relief with the wrong tone curve.

```
Create a 10-block ramp:
white, 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, black

Run it with your planned 3D Sliced settings.
Measure or inspect depth and finish.
Adjust your source depth-map curve so the useful tones land where the metal actually cuts well.
```

Suggested first calibration run: 2000 mm/s, 92% power, 100 kHz, 150-200 ns, 0.025 mm interval, 256 slices, cleanup every 20 layers at 30% power.

11. Troubleshooting

Problem	Likely cause	Fix
Relief looks flat	Depth map has weak tonal separation or too few effective passes.	Increase contrast, lower gamma, reserve darker tones for important recesses, or increase slices.
Engrave is muddy	Background/noise is being engraved; too much of the map is mid-to-dark gray.	Flatten background, brighten shallow areas, smooth the map, reduce unnecessary black regions.
Surface is rough	Too much heat or debris buildup.	Use more cleanup passes, reduce power per pass, increase speed, or use more total passes at gentler settings.
Detail disappears	Line interval too coarse, map too soft, or engraving size too small.	Use 0.020-0.025 interval, sharpen only major edges, and simplify tiny details.
Runtime is excessive	Too many dark pixels, too many	Whiten non-critical zones, reduce

	slices, overly high DPI/low interval.	slices, crop tighter, or use 128-pass preview first.
Depth is reversed	Image polarity is wrong.	Invert source image or use LightBurn Negative Image.

12. Pre-Engrave Checklist

- Confirm lens calibration and field correction.
- Clean lens and protective window.
- Verify focus on the actual metal surface.
- Run a small grayscale ramp test on the same material.
- Confirm black/deep and white/shallow behavior in LightBurn preview.
- Check that background is shallow/white unless intentionally part of the relief.
- Start with 128 or 256 slices before committing to a 512-slice final.
- Use exhaust and appropriate laser safety procedures for metal fumes and reflections.

13. Quick Recipes

13.1 Brass coin first attempt

```
LightBurn: Image -> 3D Sliced
Speed: 2000 mm/s
Power: 92%
Frequency: 100 kHz
Pulse width: 150-200 ns
Line interval: 0.025 mm
Passes/slices: 256
Angle increment: 45 or 60 degrees
Cleanup: every 20 layers, 30% power, 100 kHz
```

13.2 ZoeDepth map shaping

```
Depth clip: 5-95 percentile
Invert: yes, if black should be deepest in LightBurn
Gamma: 0.65-0.85
Background: push to near-white
Noise reduction: light bilateral smoothing
Manual edits: repaint face/object planes, fix hair/text/edges
Export: grayscale PNG for LightBurn, keep master copy
```

14. Sources and Useful References

These sources informed the workflow and confirm how LightBurn 3D Sliced mode and MOPA/fiber metal engraving are being used in practice.

1. LightBurn documentation - Galvo-specific cut settings and 3D Sliced behavior: https://docs.lightburnsoftware.com/latest/Reference/CutSettingsEditor/GalvoSpecificCutSettings/

2. LightBurn documentation - 3D Sliced engravings guide: https://docs.lightburnsoftware.com/latest/Guides/3DSlicedImage/

3. OMTech help center - fiber lasers as the preferred metal engraving technology: https://help.omtechlaser.com/hc/en-us/articles/33745233669657-5-Different-Types-of-Laser-Engravers-in-2024

4. OMTech knowledge base - metal laser marking spray for CO2 context: https://omtechlaser.com/blogs/knowledge/omtech-metal-laser-marking-spray-for-co2-laser-engravers

5. LightBurn forum - making depth maps and using 3D Slice on brass coins: https://forum.lightburnsoftware.com/t/making-depth-maps-and-using-3d-slice-on-brass-coins/142933

6. LightBurn forum - 3D Sliced depth map limitations and pass-count discussion: https://forum.lightburnsoftware.com/t/3d-sliced-depth-map-images-lightburn-limitations-behavior/188533

7. Reddit r/Laserengraving - image to depth map to engraved coin using 60W MOPA and LightBurn 3D Slice: https://www.reddit.com/r/Laserengraving/comments/1qpp4ug/image_to_depth_map_to_engraved_coin/

8. Reddit r/Laserengraving - community depth-map workflow discussion: https://www.reddit.com/r/Laserengraving/comments/1cry50i/a_new_design_i_made_yesterday_and_cut_into_a/

9. ZoeDepth repository: https://github.com/isl-org/ZoeDepth

Final recommendation: For a first serious metal relief test, use brass, a tightly cropped and hand-cleaned ZoeDepth map, 256 slices, 0.025 mm interval, about 2000 mm/s, about 92% power, 100 kHz, 150-200 ns, and cleanup every 20 layers. Move to 512 slices only after the map is visually and physically proven.